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ExpiryIQ

AI-Powered Smart Family Life Assistant

Agenda

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Quick Recap

Indian households manage dozens of recurring, time-critical responsibilities with no unified system.

ExpiryIQ is a unified, AI-powered platform that helps Indian families track and get reminded about everything with an expiry or due date — medicines, documents, insurance, bills, groceries, appliances, and family events — all in one dashboard.



Progress Since Last Review

Since our last review, we moved from a largely designed system to a functionally working application:

- Full authentication system with OTP verification
- Multi-profile family account system (JioHotstar-style)
- All 10 modules connected to a live MongoDB Atlas database
- OCR-based AI scanning
- In-app and browser push notification system
- User-configurable reminder timing per item
- AI Assistant answering from real account data

Notification System Overview



1. In-App Notifications

A notification bell in the dashboard shows real-time alerts with an unread count, pulled from a backend-scheduled daily check across all modules.

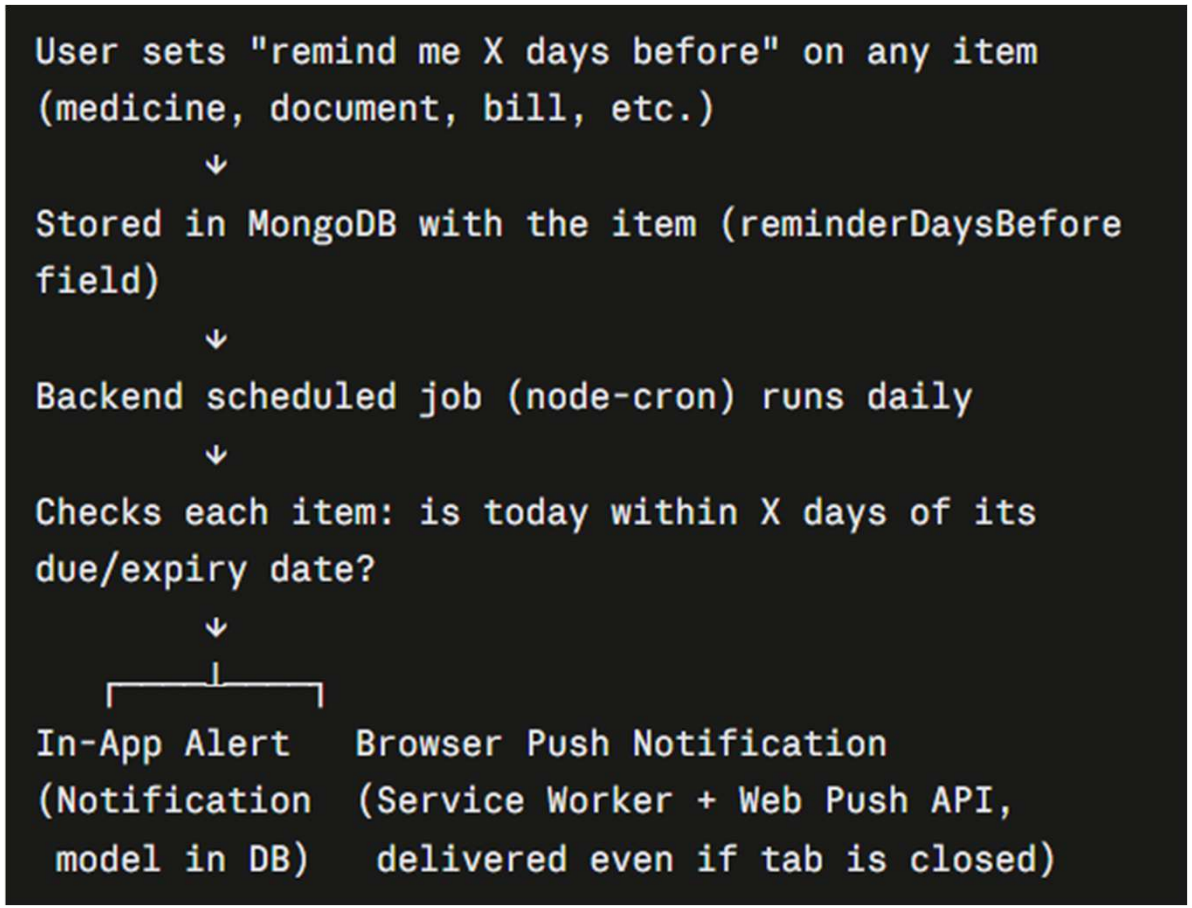
2. Browser Push Notifications

Using the Web Push API, users receive real OS-level notifications even when the ExpiryIQ tab or browser is closed — powered by a Service Worker running in the background and VAPID-based secure push authentication.

3. User-Configurable Reminder Timing

Instead of a fixed "remind me 7 days before" rule for everyone, users can now set their own preferred reminder window per item — e.g., "remind me 3 days before this bill is due" or "remind me 30 days before this policy renews." The system respects each user's chosen timing rather than a one-size-fits-all default.

How Notifications Actually Work



OTP-Based Authentication Flow

Account creation security:

- When a user registers a new family account, an OTP is sent to verify their email before the account is activated
- This confirms the account owner has genuine access to the registered email, preventing fake or mistyped-email signups

Family member addition security:

- When a new family member profile is added under an existing account, an OTP is also sent — to the account owner, not the new member
- This ensures only the verified account owner can authorize new profiles being created under their family account, preventing unauthorized additions

Technology Used for Notifications

COMPONENT	TECHNOLOGY
Scheduled daily checks	node-cron
In-app notification storage	MongoDB (Notification model)
Browser push delivery	Web Push API + Service Worker
Push authentication	VAPID keys (web-push npm package)
OTP generation & verification	Custom backend logic + email delivery

Challenges Faced

- Implementing OTP verification correctly for two different flows (account creation vs. family member addition) without confusing which email receives the code
- Configuring the Service Worker and VAPID push authentication correctly — required careful setup of file paths, permissions, and key generation
- Generalizing the notification scheduler to respect each user's own custom reminder timing per item, rather than one fixed rule for everyone.

Current Status

- Authentication, OTP, and multi-profile system — complete
- All 10 modules connected to live data
- OCR AI scanning — working on MediTrack and DocuVault
- Notification system (in-app + browser push + configurable timing) — complete

Next Steps

- 1. Direct Purchase Integration
- Add Amazon/Flipkart buy links to medicines and pantry items
- When stock is low or close to expiry, user can click a "Reorder Now" button that opens the product directly on the retailer's site
- Removes the manual step of searching for the same item again — closes the loop from "alert" to "action"
- 2. Spending Analytics Dashboard
- Visual graphs showing total money spent across the household
- Breakdown by module — medicines, bills, insurance premiums, pantry restocking — so users can see where most of their spending goes
- Helps families track and plan monthly household expenses, not just track expiry dates.

Conclusion

ExpiryIQ transforms household management from reactive and fragmented into proactive and unified.

By combining automation, prediction, and cultural relevance, it addresses a real, everyday problem faced by millions of Indian families — one that no existing application has fully solved.

"Because every household deadline deserves the same intelligence as a work deadline."